crash program is what occurs when important elements of the research personnel are suddenly pulled from the proj-ects on which they have been working and concentrated on some new task which may have great importance at the moment but which, fre-quently, is not worth all the disruption it causes. The essence of successful commercial research is that only tasks be selected which promise to give dollar rewards of many times the cost of the research. However, once a project is started, to allow budget considerations and other extra-neous factors outside the project itself to curtail or accelerate it invari-ably expands the total cost in relation to the benefits obtained.

Some top managements do not seem to understand this. I have heard executives of small but successful electronic companies express surprisingly little fear of the competition of one of the giants of the industry. This lack of worry concerning the ability of the much larger company to produce competitive products is not due to lack of respect for the capabilities of the larger company's individual researchers or unawareness of what might otherwise be accomplished with the large sums the big company regularly spends on research. Rather it is the historic tendency of this larger company to interrupt regular research proj-ects with crash programs to attain the immediate goals of top management that has produced this feeling. Similarly, some years ago I heard that while they desired no publicity on the matter for obvious reasons, an outstanding technical college quietly advised its graduating class to avoid employment with a certain oil company. This was because top management of that company had a tendency to hire highly skilled peo-ple for what would normally be about five-year projects. Then in about three years the company would lose interest in the particular project and abandon it, thereby not only wasting their own money but preventing those employed from gaining the technical reputation for accomplishment that otherwise might have come to them.

Another factor making proper investment evaluation of research even more complex is how to evaluate the large amount of research related to defense contracts. A great deal of such research is frequently done not at the expense of the company doing it, but for the account of the federal government. Some of the subcontractors in the defense field also do significant research for the account of the contractors whom they are supplying. Should such totals be appraised by the investor as being as significant as research done at a company's own expense? If not, how should it be valued in relation to